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# Bayesian spatial-temporal analysis and determinants of cardiovascular diseases in Tanzania mainland

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## Abstract

**Background** Cardiovascular Diseases (CVDs) are health-threatening conditions that account for high mortality in the world. Approximately 23.6 million deaths due to CVD is expected in the year 2030 worldwide. The CVD burden is more severe in developing countries, including Tanzania.

**Objectives** This study analyzed the spatial-temporal trends and determinants of cardiovascular diseases in Tanzania from 2010 to 2019.

**Methods** Individual data were extracted from Jakaya Kikwete Cardiac Institute (JKCI), Mbeya Zonal Referral Hospital (MZRH), Kilimanjaro Christian Medical Centre (KCMC) and Bugando hospitals and the geographical data from TMA. The model containing spatial and temporal components was analyzed using the Bayesian hierarchical method implemented using Integrated Nested Laplace Approximation (INLA).

**Results** The results found that the incidence of CVD increased from 2010 to 2014 and decreased from 2015 to 2019. The southern highlands, lake, central and coastal zones were more likely to have CVD problems than others. It was also revealed that people aged 60–64 years OR = 1.49, females OR = 1.51, smokers OR = 1.76, alcohol drinkers OR = 1.48, and overweight OR = 1.89 were more likely to have CVD problems. Additionally, a 1$^{o}$C increase in the average annual air maximum temperature was related to a 14% risk of developing CVD problems. The study revealed that the model, which included spatial and temporal random effects, was the best-predicting model.

**Conclusion** The study shows a decreased CVD incidence rate from 2015 to 2019. The CVD incidences occurred more in Tanzania's coastal and lake areas between 2010 and 2019. The demographic, lifestyle and geographical risk factors were significantly associated with the CVD.

**Keywords** Cardiovascular diseases (CVD), Unstructured random effects, Structured random effects, Bayesian spatial-temporal, INLA

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## Introduction

Cardiovascular diseases (CVDs) are health-threatening conditions that have high incidence and mortality rates worldwide [1]. In the year 2030, CVD deaths are expected to reach approximately 23.6 million worldwide [1, 2]. The CVD burden is more severe in developing countries, as CVD accounts for almost 60% of the global disease burden [1], and three-quarters of CVD-related deaths occur in these countries [2]. The disease no longer affects only older people; young people are also affected by it [2] because they are vulnerable to CVD risk factors [3]. CVD-related deaths accounted for 13% of the total number of NCD-related deaths in Tanzania [4]. Despite different measures taken to combat CVD problems in developing countries such as Tanzania, the problem is increasing.

Assessing the spatial and temporal risk factors related to CVD incidence in the population is vital to informing appropriate authorities about well-informed decisions. One study [5] used geographically weighted regression (GWR) and ordinary least squares (OLS) methods to model the spatial association between social deprivation and CVD mortality in the United States of America and found that GWR (R2=65%) performed better than OLS (R2=57%). Another study used varying coefficient (SVC) and GWR models to analyze the varying distributions of people living with diabetes and hypertension and found that SVC was superior to GWR because it is suitable for significant volumes of data compared to GWR [6]. Another study [7] investigated the spatial distribution of cardiovascular disease mortality risk factors using OLS (nonspatial), spatial lag and spatial error models. The results of the two spatial models (spatial lag and spatial error) were compared with those of OLS using the likelihood ratio (LR) test. The results showed that the spatial error model (LR=64.18) performed better than the spatial lag model (LR=16.03), and all the models performed better than the OLS model. Spatial models performed better than nonspatial models in analyzing the associations among CVD risk factors.

However, Bayesian analysis, which includes spatial and spatial-temporal structures, has been used in epidemiology data [6]. It allows estimating and predicting spatial patterns and relationships between variables while incorporating prior knowledge and uncertainty. Bayesian spatial modelling is flexible enough to incorporate spatial dependence, uncertainty, and prior information into the analysis. It also allows the estimation of model parameters and predicts values of the unobserved locations using the observed ones. The Bayesian framework estimates the unknown parameters based on the posterior distribution given the observed data and the prior distribution [8]. Bayesian analysis drives the model parameters based on the likelihood and prior probability distributions. Studying the spatial-temporal relationship between the risk of developing CVD and associated risk factors is essential for disease management and decision-making [6]. In recent years, different researchers have explored the temporal and spatial patterns and related risk factors for CVD.

Spatial-temporal studies such as [9] have investigated how CVD mortality relates to PM2.5 air pollution exposure. [10] examined the spatial-temporal distribution of ischemic heart disease (IHD) in Manitoba province in Canada and investigated the association between area-level socioeconomic factors and disease incidence. Another study by [11] examined spatiotemporal trends and the effect of nighttime light on CVD mortality in China. According to different spatial-temporal analyses of CVD and its associated risk factors, no study has used demographic, lifestyle and geographical factors to analyze the joint association with CVD incidence. Knowing the spatial and temporal trends of CVD incidence and the joint variables is important because these variables correlate. Therefore, this paper assesses the spatial-temporal trends and demographic, lifestyle and geographical exposure of CVD incidence.

## Materials and methods

### Data

The dataset consists of 3553 observations extracted from the records departments of the designated hospitals from 2010 to 2019. Referral hospitals in Tanzania receive any individual seeking health services to the hospital regardless of whether they are referred from other hospitals. A list of outpatients who visited cardiac units in four selected hospitals for the first time between 2010 and 2019 was sampled using a simple random sampling technique. People between 25 and 64 years old during their first visit to the cardiac unit were considered in this study. The sampled patient's file was used to obtain the data recorded by a doctor. The hospitals Jakaya Kikwete Cardiac Institute (JKCI), Mbeya Zonal Referral Hospital (MZRH), Kilimanjaro Christian Medical Centre (KCMC) and Bugando are located in Dar es Salaam, Mbeya, Kilimanjaro and Mwanza regions respectively. These hospitals were purposively selected because they receive many patients from their respective zones, which increases the probability of data availability for the disease condition under study. These hospitals are monitored and supported by the government of Tanzania to ensure the availability of quality infrastructure [12]. Additionally, geographical data were collected from the Tanzania Meteorological Agency (TMA). The data from selected hospitals and TMA were combined using the region of residence as a common attribute to form a dataset of 3553 observations.

## Variables of the study

### *Outcome variable*

Outpatients who were admitted to the cardiac unit for the first time between 2010 and 2019 in the selected hospitals were diagnosed with CVD incidences and were assigned (1) if an individual had a CVD problem or (0) if not. The CVDs with their respective International Classification of Diseases (ICD-10) codes were ischemic heart disease (*I20-I25*), acute myocardial infarction (I*22*), stroke (*I60-I69*), rheumatic heart disease (RHD), congestive heart failure (*I50.1-I50.3*), hypertensive heart disease (*I10*) and coronary artery disease (CAD). These diseases were selected because they are the leading cause of morbidity and mortality.

### *Covariates*

Covariates included in this study were selected from other studies for which they have determined their association with cardiovascular diseases. Factors associated with CVD that are considered in this study are demographic [13], lifestyle [14], and geographical [15] factors. These covariates are the individual patient's sex, age, body mass index (BMI), systolic blood pressure (SBP), diastolic blood pressure (DBP), smoking status and alcohol consumption habits. The SBP>130 mmHg and DBP>80 mmHg indicated hypertension [16]. Also, geographical covariates were annual average air humidity, air temperature, and education status of an area based on primary-level education of the region, proxied by grade point average (GPA). The individual data were merged with the geographical data using the region of residence for a patient as a link attribute.

## Model specification

The spatial-temporal association between CVD and the covariates was modeled using a generalized linear mixed model (GLMM). The probability of randomly selected individuals following a Bernoulli distribution is expressed as $y_{it} \sim Bern(\theta_{it})$, where $\theta_{it}$ is the probability of an individual having CVD incidence in the $i^{th}$ region in year $t$ with $E(Y_{it}) = \theta_{it}$ and $Var(Y_{it}) = \theta_{it}(1-\theta_{it})$. The GLMM included random terms $f(.)$ and links the linear predictors and the mean of the response variable using the logit link function expressed as:

$$logit(\theta_{it}) = log\left(\frac{\theta_{it}}{1-\theta_{it}}\right) = \beta_0 + \sum_{k=1}^{k=p} \beta_{ki} X_{kit} + f(.)$$

where $\theta_{it}$ denotes the probability of an individual having CVD incidence, $\beta_0$ denotes the model's intercept, $X_{kijt}$ is the $k$ covariate for individual $j$ in region $i$ during year $t$ and $\beta_{ki}$ are regression coefficients for the observed covariates, and $f(.)$ represents the random effects. The $p$ is the total number of covariates used in this study.

## Spatial-temporal model specifications and parameter estimation

Three models were adopted for this study: a first model with spatial random effects ($M_1$), the second with temporal effects ($M_2$) and the third with spatial-temporal effects ($M_3$). In the first model, $M_1$, it considered covariates and unobserved covariates (unstructured or structured spatial effects), which might significantly influence CVD incidence. Unstructured spatial effects$(u_i)$ are unobserved covariates inherent within a region that cause correlations among individuals in the same region. The correlation between individuals within the region may exist due to the commonalities of practices such as the same lifestyle, culture, climate conditions, and health intervention. On the other hand, structured spatial effects $(s_i)$ are other unobserved covariates that vary spatially across regions, resulting in spatial autocorrelation due to geographical proximity. Considering only unstructured or structured random terms results in biased model parameter estimates [13]. Therefore, model $M_1$ accounts for unstructured $(u_i)$ and structured $(s_i)$ random effects.

$$M_1 : log\left(\frac{\theta_{it}}{1-\theta_{it}}\right) = \beta_0 + \sum_{k=1}^{k=11} \beta_{ki} X_{kit} + u_i + s_i$$

The unstructured spatial effect was assumed to be independent and identically distributed with mean zero and variance $\sigma_u^2$. The structured random effects were modeled using Besag models. The Besag model uses conditionally autoregressive (CAR) models to estimate spatial correlation for disease mapping. The CAR prior models consider that the correlated random effects in region $\boldsymbol{i}$ $(V_i)$ given the correlated random effect in region $j$ $(V_j)$ are normally distributed with mean $\frac{\sum_{j\in R_i} W_{ij} V_j}{\sum_{j\in R_i} W_{ij}}$ and variance $\frac{\sigma_V^2}{\sum_{j\in R_i} W_{ij}}$, where $R_i$ is the total number of regions that share the same boundaries with region $i$ and $W_{ij}$ is the spatial neighbor structure such that $W_{ij} = 1$ if regions $i$ and $j$ share the same boundary and zero otherwise.

The second model $(M_2)$ considered the covariates and temporal random effects. The temporal effect was assumed to be nonlinear; hence, a nonparametric approach was implemented based on [17]. The unstructured $(\gamma_t)$ temporal effect was considered to be independent and identically distributed $(iid)$ with a mean and variance of zero and $\sigma_\gamma^2$, respectively. Additionally, the structured $(\nu_t)$ temporal random effect was modeled using the autoregressive first-order AR(1), first-order (RW(1)) and second-order (RW(2)) random walks. The best model among those mentioned above was later included in spatial-temporal modeling.

$$M_2 : \ log\left(\frac{\theta_{ij}}{1-\theta_{ij}}\right) = \beta_0 + \sum_{k=1}^{k=11}\beta_{ki}X_{kij} + \gamma_t + \nu_t$$

The third model ($M_3$) combined the spatial model ($M_1$) and the best temporal model ($M_2$) random components, and the fourth ($M_4$) model added space-time interaction ($\delta_{it}$) on $M_3$

$$M_3 : \ log\left(\frac{\theta_{ij}}{1-\theta_{ij}}\right) = \beta_0 + \sum_{k=1}^{k=11}\beta_{ki}X_{kij} + u_i + s_i + \gamma_t + \nu_t$$

$$M_4 : log\left(\frac{\theta_{ij}}{1-\theta_{ij}}\right) = \beta_0 + \sum_{k=1}^{k=11}\beta_{ki}X_{kij} + u_i + s_i + \gamma_t + \nu_t + \delta_{it}$$

The R-INLA package was used to estimate the parameters of the model $\theta = \{\beta, \upsilon, s, \gamma, \nu, \delta\}$ and hyperparameters $\psi = \{\tau_\upsilon, \tau_s, \tau_\gamma, \tau_\nu\}$, and the gamma prior was used. The R-INLA package in R software is accessible through http://www.r-inla.org.

### Model selection

This study considered three models, as described in Sect. 2.3. Selection methods, namely, the Deviance Information Criteria (DIC) [18, 19], Watanabe Akaike information criteria (WAIC) [2] and conditional predictive ordinate (CPO) [20], were used to compare the best fits of the three models. The DIC is obtained by multiplying the posterior mean of the deviance by two and subtracting the deviance of the posterior mean values for the model parameters. These methods were applied during model fitting on the training dataset. The model with the highest CPO and lowest DIC and WAIC was chosen as the best for predicting CVD.

The sample data obtained from selected hospitals were weighted to represent the entire population of Tanzania's mainland. The weights were calculated for each hospital and assigned for each respective observation. It was so done because the sample for each hospital was different. Later the weights were incorporated into the sample data and analyzed using the INLA weight function.

### Priors sensitivity analysis

The sensitivity analysis was performed to find the best hyperparameter priors for modelling CVD incidences using the best model identified. Different priors were selected for inverse gamma distribution to measure the robustness of the posterior mean distribution. The distribution of the priors determines whether the posterior mean changes or not. The hyperparameters chosen for the inverse gamma distribution were (a=b=0.01), (a=0.5 and b=0.0005) and (a=1 and b=0.026) based on [21]. The best hyperparameter was selected based on the lowest DIC value [22].

**Table 1** The descriptive statistics for geographical factors

| Variable | Mean | Std. Dev. | Min | Max |
|---|---|---|---|---|
| Humidity | 71.10 | 7.21 | 49 | 85 |
| MaxTemp | 28.47 | 2.70 | 18.6 | 32.2 |
| GPA | 3.12 | 0.33 | 2.38 | 3.88 |

### Model assessment

The CVD dataset was imbalanced; hence, the area under the ROC curve (AUC) was used to identify the model with overall good predictive power. The receiver operating characteristic (ROC) curve demonstrates the area under the curve (AUC) by plotting the true positive rate (TPR) against the false positive rate (FPR) [23]. It measures the model's ability to predict TRUE or FALSE classes correctly. The sensitivity or TPR measures the percentage of people with CVD cases, and the specificity (FPR) measures the percentage of individuals correctly identified as not having CVD. The model performs better when the AUC approaches the ROC curve's top-left corner and is less accurate when it lies along the diagonal (FPR=TPR) [24]. The model with the highest AUC was the best prediction model.

## Results

This study intended to analyze the spatial-temporal trend and determinants of CVD incidence in mainland Tanzania using data from four tertiary hospitals and the TMA. Descriptive statistics are also presented in this section.

### Analysis of descriptive statistics

#### *Geographical factors*

The result revealed that the maximum and minimum humidity was 49 and 85, respectively with an annual average of 71.1mmHg for ten years. The MaxTemp had a minimum value of 18.6$^{\circ}$C and a maximum value of 32.2$^0$C with an average yearly MaxTemp of 28.5$^0$C. The region's minimum and maximum academic performance had a GPA of 2.38 and 3.88, respectively (Table 1).

#### *Lifestyle factors*

The lifestyle factors revealed that smokers (839) and alcohol drinkers (977) were fewer than non-smokers and non-drinkers (Figs. 1 and 2), respectively.

### Cardiovascular disease incidence in Tanzania

The crude CVD incidence rate for each region per 100,000 population at risk is presented in Fig. 3. The descriptive statistics show that the Northern and Southern Highlands zones had the highest crude CVD incidence rate. The central, southern and western zones have a low crude CVD incidence rate of less than 12.4 per 100,000 population for ten years (2010–2019) (Fig. 3).

The adjusted CVD incidence rates (ACIR) for each region from 2010 to 2019 revealed decreased disease incidence rates for ten years. It is shown that some areas had ACIRs ranging between 6.8 and 8.5 per 100,000 people at risk between 2010 and 2012. From 2016 to 2019,

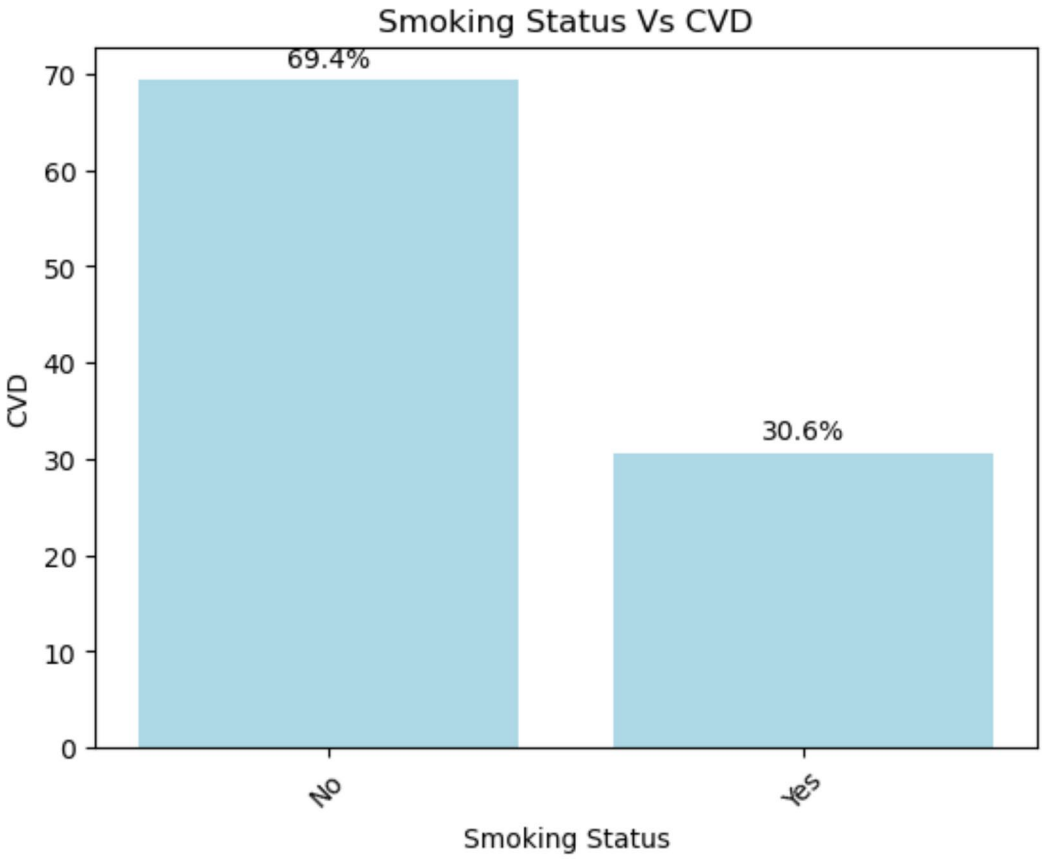



**Fig. 1** Distribution of smoking habit and CVD incidence

the ACIR decreased to below 5.1 per 100,000 population. In a year, most regions experienced an ACIR of less than 1.7 per 100,000 population (Fig. 4).

#### Model fit performance comparison

The DIC, WAIC and CPO for the three models are presented in Table 2. The model with the lowest DIC and WAIC and highest CPO performed best. Model ($M_3$) with covariates and temporal and spatial random effects was the best-fitted model. Model ($M_3$) has the lowest DIC (16008.28) and WAIC (16398.89) and the highest CPO (987.11), followed by $M_4$ with covariates and spatial effects DIC (16053.91), WAIC (16431.09) and CPO (978.35). Models that included only covariates ($M_1$) and temporal effects ($M_2$)were the worst models. Based on this result, a model with temporal and spatial random terms exhibited increased model performance.

#### Prior sensitivity analysis

The DIC for each selected prior was almost the same, indicating the robust posterior distribution as shown in Table 3. The DIC values for the selected priors are closely similar, with varying random effects. The hyperparameter prior information for inverse gamma a=0.01 and b=0.01 had the lowest DIC (16007.38); therefore, it was implemented to model the CVD incidences.

#### Model testing and validation using AUC

The prediction performance of the three models using the AUC revealed that $M_3$ with covariates and spatial and temporal random effects, best predicted CVD outcomes with the highest AUC (0.6668), followed by the $M_1$ AUC (0.6624), and the worst model was the $M_2$ AUC (0. 6361), as shown in Fig. 5. Based on this result, including spatial and temporal random terms provides the best predictions. Therefore, the results and other interpretations are based on the model ($M_3$), which included covariates of spatial and temporal random terms.

#### Relationships between CVD incidence and risk factors

The risk factors associated with CVD incidence were assessed using $M_3$ with covariates and spatial and temporal random components. The risk factors included in the models were demographic (sex and age), lifestyle (smoking, alcohol consumption, BMI and BP) and geographical (average annual maximum temperature (MaxTemp), average yearly humidity and region's education status

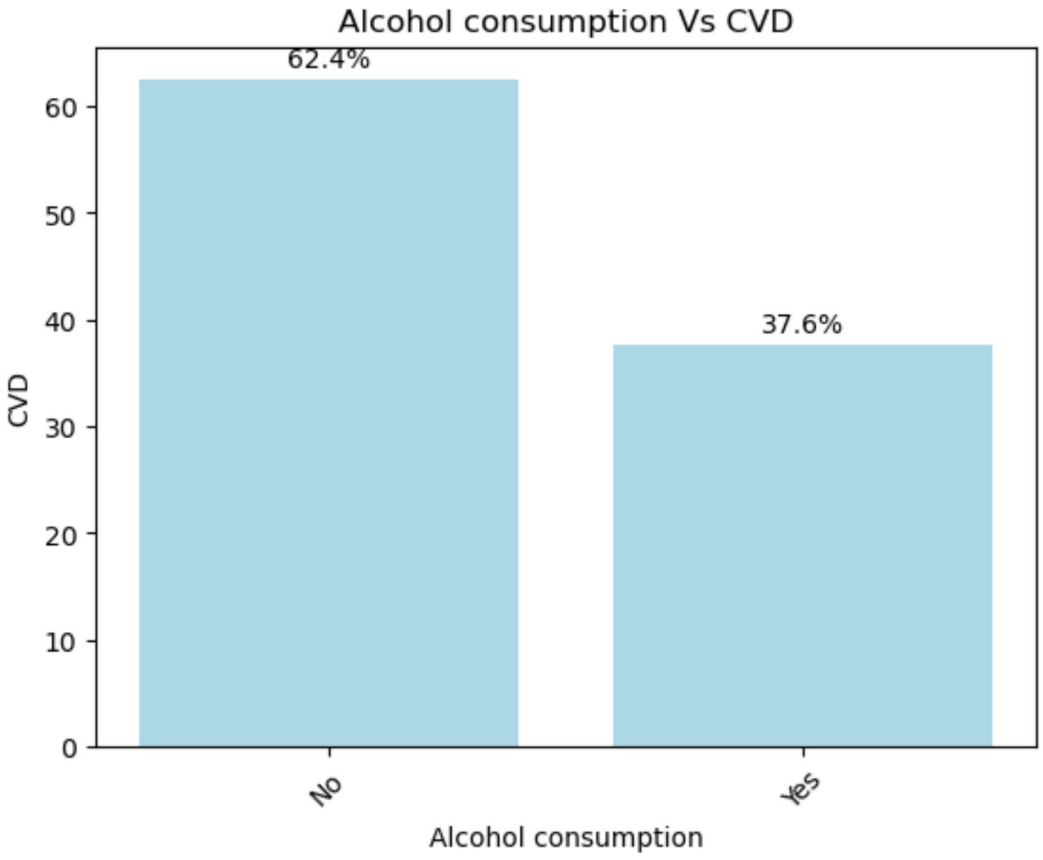



**Fig. 2** Alcohol consumption and CVD incidences

(GPA)). All these covariates were statistically significant at the 90% confidence interval.

The results showed that older people aged between 60 to 64 years had greater odds of having CVD than people aged 44 to 59 years with odds ratios (ORs) of OR = 1.08 (95% CI: 0.86–1.32). Females (OR = 1.51, 95% CI: 1.26–1.81), smokers (OR = 1.76, 95% CI: 1.44–2.14), alcohol drinkers ([OR = 1.48, 95% CI: 1.23–1.77) and overweight individuals (OR = 1.89, 95% CI: 1.51–2.36]) were more likely to have CVD problems than their counterparts. The results also revealed that a 1 °C increase in the average annual air maximum temperature (OR = 1.14, 95% CI: 1.06–1.23) was related to a 14% risk of developing CVD problems (Table 4).

#### Spatial random effects

The unstructured $(u_i)$ and structured $(s_i)$ random effects revealed that CVD incidence is spatially distributed across the country (Figs. 6 and 7, respectively). The darker blue represents regions with a higher risk of CVD for individuals, and the lighter blue represents regions with the lowest likelihood of having CVD. The southern highlands, lake, central and coastal zones are more likely to have CVD problems than other regions, as shown in Figs. 6 and 7. Areas with negative spatial random effects indicate that the CVD incidence rate is less than expected; otherwise, there are positive spatial effects.

#### Temporal distribution of CVD

Figure 8 shows the temporal trend of CVD relative risk in the Tanzania mainland. The study revealed that CVD incidence increased beginning in 2010, reached its highest rate in 2014, and then decreased from 2015 to 2019, reaching its lowest rate in 2017. The overall risk of CVD incidence showed a declining trend from 2010 to 2019. Generally, the relative risk of a temporal effect of CVD was high (AOR > 1) from 2010 to 2014, and from 2015 to 2019, there was less risk of CVD incidence.

#### Spatial-temporal distribution analysis

The study revealed the spatial-temporal varying pattern of CVD incidence, as shown in Fig. 9. CVD incidences were greater in the lake zone regions, central zone, coastal zone and northern zone of the Tanzania mainland for the entire period (2010–2019). Nevertheless, regions comprising Tanzania's southern highlands and northern zone had low CVD incidence rates for 2010–2019.

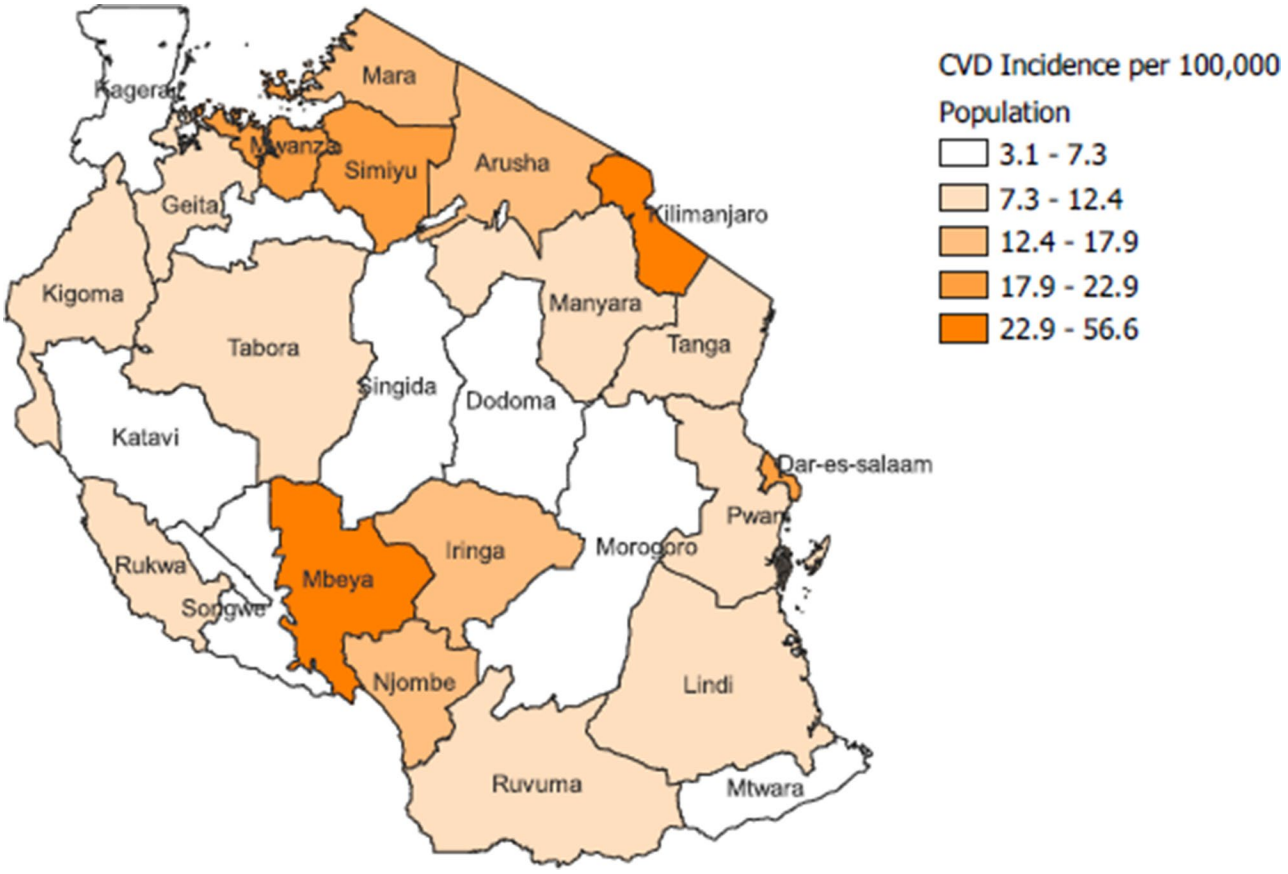



**Fig. 3** The distribution of crude CVD incidence rate for ten years (2010–2019) in Tanzania

## Discussion

This paper assessed the spatial-temporal distribution of CVD and the effects of individual and geographical-level factors associated with CVD incidence in Tanzania using data collected from four tertiary hospitals and the TMA from 2010 to 2019. This study examined the spatial-temporal variation in CVD incidence to identify spatial-temporal areas with a high risk of CVD. The performances of the three models were compared to determine the best model that best fit the CVD dataset. The study revealed that a model that included spatial and temporal random effects was best for predicting CVD incidence. In this regard, including spatial and temporal effects in predicting the risk of CVD incidence is essential for ensuring people's healthy lives. This result is consistent with a study by [25, 26], which found that considering spatial and temporal random effects in modeling CVD problems is highly advantageous. Other studies by [6, 9, 10, 27] used Bayesian spatial-temporal models for the data collected over space and time to study the association of risk factors and CVD mortalities or incidences.

The incidence of CVD increased from 2010 to 2014 and decreased from 2015 to 2019 in the study area. This result is similar to that of a study by [28], who reported a decreasing trend in CVD incidence from 1996 to 2017 worldwide. A possible explanation for the decrease in CVD incidence from 2015 to 2019 could be the implementation of the government's 2009–2015 plan for reducing the incidence of NCDs, including CVD [3]. People are also aware of the risk factors related to CVD, so they take necessary precautions [29]. The government of Tanzania has recently invested in improving health infrastructures to strengthen health service provision to citizens for better lives compared to other times [30].

The study revealed that CVD incidence was higher in the regions located in the lake, central and coastal zones of the Tanzania mainland over ten years (2010–2019). This finding is the same as that of a study by [31], which revealed that people living near coastal areas have a 24% greater risk of myocardial infarction (MI) than people living far away. This observation could be due to these regions having high income per capita, represented by their higher GDP [32]. The growth domestic product (GDP) is high in the lake zone, contributing to 25.9% of the total GDP of the country, followed by that of Dar es Salaam (17.3%), according to the BOT report of 2018/2019 [33], which may influence CVD problems. A higher GDP indicates increased wealth associated with

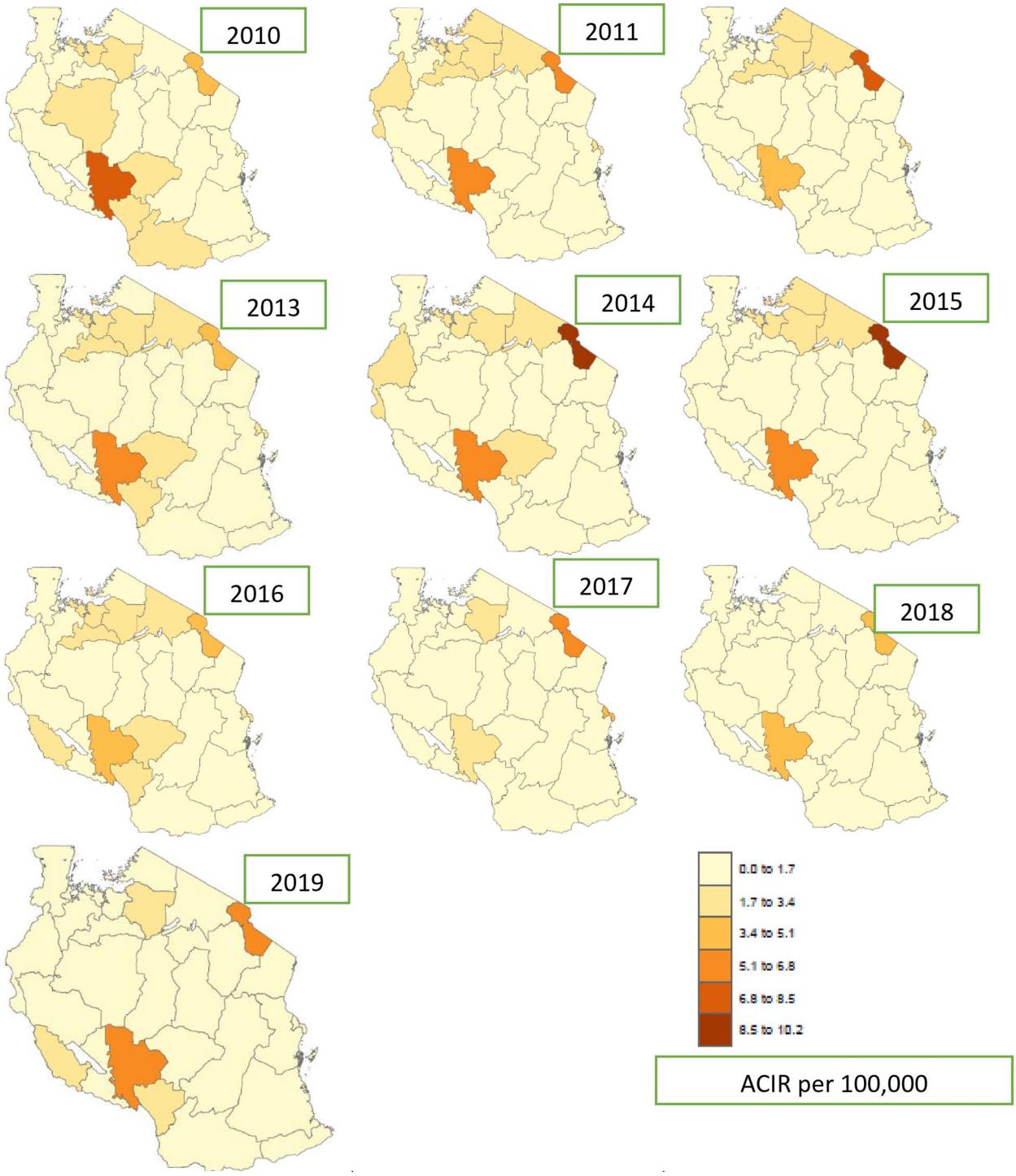



**Fig. 4** Annual adjusted CVD incidence rate for each region in Tanzania from 2010 to 2019

CVD problems [34, 35]. Regions such as Dar es Salaam, Tanga, Mwanza and Dodoma are urban areas where it is expected to have a high risk of CVD problems [36] due to increased use of processed foods, sedentary lives and air pollution due to industrialization and overcrowding.

This study revealed that females have greater odds of developing CVD than males. This result is in line with other studies [35, 37, 38] reporting that women are at greater risk of CVD than men are. This may be the result of hormonal impairment in women, particularly during

**Table 2** Comparison of the model fit performance results

| Model Description | Model | DIC | WAIC | CPO | Decision |
|---|---|---|---|---|---|
| Spatial random effects | M1 | 16171.64 | 16500.32 | 972.49 | |
| Temporal random effects | M2.1 (ar1) | 16748.65 | 16948.02 | 875.43 | |
| | M2.2 (rw1) | 16748.84 | 16946.44 | 875.48 | |
| | M2.3 (rw2) | 16745.85 | 16943.94 | 875.56 | |
| **Spatial and Temporal Effects** | **M3** | **16008.28** | **16398.89** | **987.11** | ***Overall Best model*** |
| Space-Time interaction | M4 | 16053.91 | 16431.09 | 978.35 | |

the postmenopausal stage, caused by a decrease in estrogen and testosterone hormones, which regulate the cardiovascular system and hence increase the risk of CVD [37–39]. Moreover, postmenopausal side effects such as physical stress and emotions have been reported to increase women's CVD risk [38].

Smokers and alcohol drinkers have a greater chance of developing CVD than their counterparts. Heavy alcohol consumption [40] and smoking or passive smoking [41] are associated with a greater risk for coronary heart disease, stroke, and hypertension than moderate alcohol consumption and nonsmokers, respectively. Other studies revealed that a history of smoking [42] is associated with CVD, which is similar to the findings of this study. It has also been reported that overweight or obese individuals increase the risk of CVD compared to underweight and normal individuals. Other studies revealed that body overweight and obesity are associated with coronary artery disease (CAD), stroke incidence [43] and CVD mortality [14].

Similarly, this study revealed that a region's annual average temperature was positively associated with increased CVD incidence (OR = 1.14, 95% CI: 1.09–1.26). This result is similar to that of a study by [44], which revealed that a temperature increase of 1 °C is associated with a 2.1% increase in CVD morbidity and mortality. This can be due to hot weather, which makes it more difficult for the human body to maintain a normal temperature, which places excess work on the heart. Previous research has even suggested that cardiovascular disease is sensitive to weather excesses (extreme low and high) [45].

### Limitations of the study

Some limitations considered in this study have implications for the interpretations of the findings. Firstly, this study was retrospective and analyzed existing data collected from the medical records department from

**Table 3** The various hyperparameters selected for posterior mean and standard errors for the best spatial-temporal model ($M_3$)

| Spatial and temporal effects | a = 0.01, b = 0.01 | a = 0.5, b = 0.0005 | a = 1, b = 0.026 |
|---|---|---|---|
| $\sigma^2_{ustruc}$ CI: 0.025–0.975 | 0.834 (0.28) [0.398–1.49] | 0.867 (0.28) [0.425–1.52] | 0.911(0.293) [0.455-1.60] |
| $\sigma^2_{struc}$ CI: 0.025–0.975 | 22.523 (44.91) [0.525–126.62] | 850.101 (1130.67) [13.882-3895.55] | 38.389 (36.834) [3.114–136.45] |
| $\gamma^2_{Time}$ CI: 0.025–0.975 | 15.723 (11.14) [3.371–44.92] | 21.078 (14.97) [4.645–60.37] | 20.277 (12.701) [5.201–53.17] |
| DIC | 16007.38 | 16007.98 | 16008.04 |

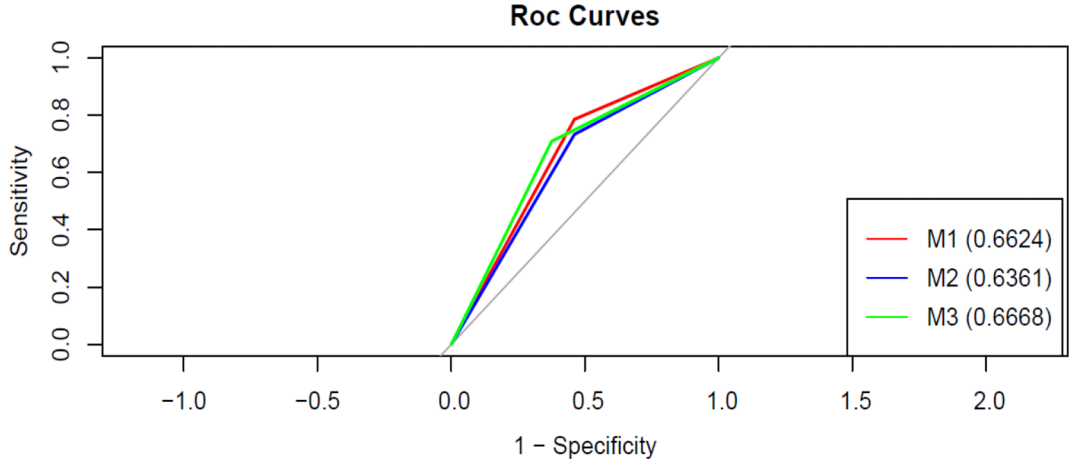



**Fig. 5** Area under the ROC curve (AUC) for model prediction performance comparison

**Table 4** Estimates of odds ratios and 95% bayesian credible intervals for $(M_3)$ for assessing risk factors associated with cardiovascular diseases

| Parameter | Posterior Mean | Odds Ratio (OR) | SD | 95% Credible Interval [CI] |
|---|---|---|---|---|
| Intercept | -0.7378 | 0.4782 | 4.15 | [0.03, 8.38] |
| ***Demographic factors*** | | | | |
| Age (Ref = Middle age 44–59) | | | | |
| Young (25–43) | 0.4027 | 1.50 | 1.16 | [1.119, 2.00] |
| Elders (60–64) | 0.0756 | 1.08 | 1.11 | [0.86, 1.32] |
| Female (Ref = Male) | 0.4155 | 1.51 | 1.10 | [1.26, 1.81] |
| ***Life style risk factors*** | | | | |
| Smoking (Ref = No) | 0.5668 | 1.76 | 1.11 | [1.44, 2.14] |
| Alcohol (Ref = No) | 0.3895 | 1.48 | 1.10 | [1.23, 1.77] |
| ***Clinically diagnosed factors*** | | | | |
| BP (Ref = Low BP) | -0.0776 | 0.92 | 1.09 | [0.78, 1.10] |
| BMI (Ref = Obesity) | | | | |
| Underweight | -1.6537 | 0.19 | 1.21 | [0.13, 0.28] |
| Normal | -1.0753 | 0.34 | 1.19 | [0.24, 0.48] |
| Overweight | 0.6382 | 1.89 | 1.12 | [1.51, 2.36] |
| ***Geographical factors*** | | | | |
| Humidity | -0.0200 | 0.98 | 1.01 | [0.96, 1.00] |
| MaxTemp | 0.1329 | 1.14 | 1.04 | [1.06, 1.23] |
| GPA | -0.2770 | 0.78 | 1.17 | [0.57, 1.07] |
| Rho for spatial component | 0.64 | | 0.25 | [0.06, 0.97] |

patients files. In their files, there were unrecorded factors that contributed to CVD problems, which doctors did not record during patient medical treatment attendance. Although other factors were not recorded, they did not affect the findings of this study because the recorded factors served the purpose.

Secondly, the data were collected from four referral hospitals inTanzania mainland. These hospitals are situated in Dar es Salaam (JKCI), Mwanza (Bugando), Moshi (KCMC) and Mbeya (MZRH) which are urban regions of Tanzania. The data collected from these hospitals is representative of the population since they are zonal-representative hospitals serving people from all regions of the respective zones. These hospitals cater to people from both urban and rural areas. This situation does not affect the findings of this study because the sample data included all areas of Tanzania mainland. For generalizability, the sample data was weighted to represent the entire population of Tanzania.

Thirdly, 365 (10.3%) were missing data for body mass index BMI variable. Other patients had missing data about their weight and height or body mass index in their files. The missed values were replaced by imputed values to avoid biased estimates and misleading interpretations of the results. The k-Nearest Neighbor (kNN) was used to impute missing values because it can handle numerical and categorical data simultaneously and has the lowest absolute mean difference compared to other methods [46].

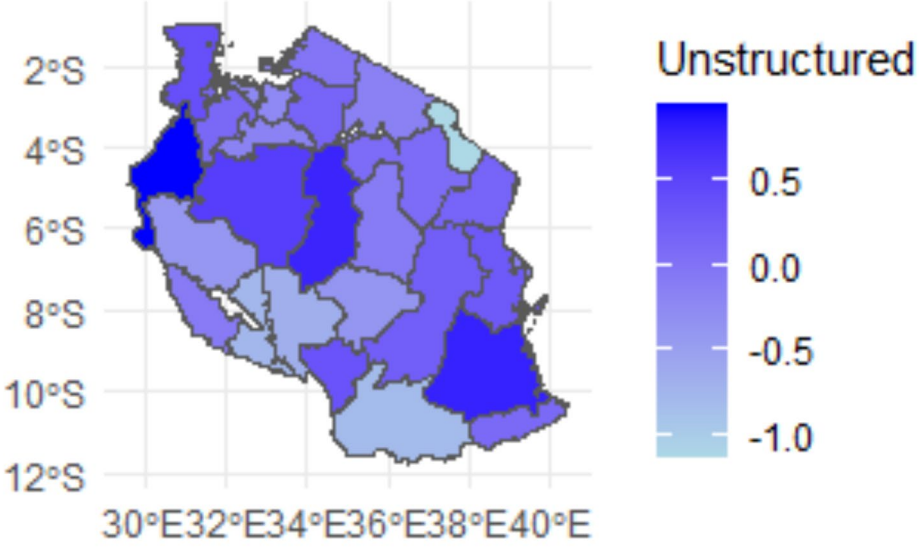



**Fig. 6** Unstructured spatial variation of CVD posterior estimates in Tanzania mainland

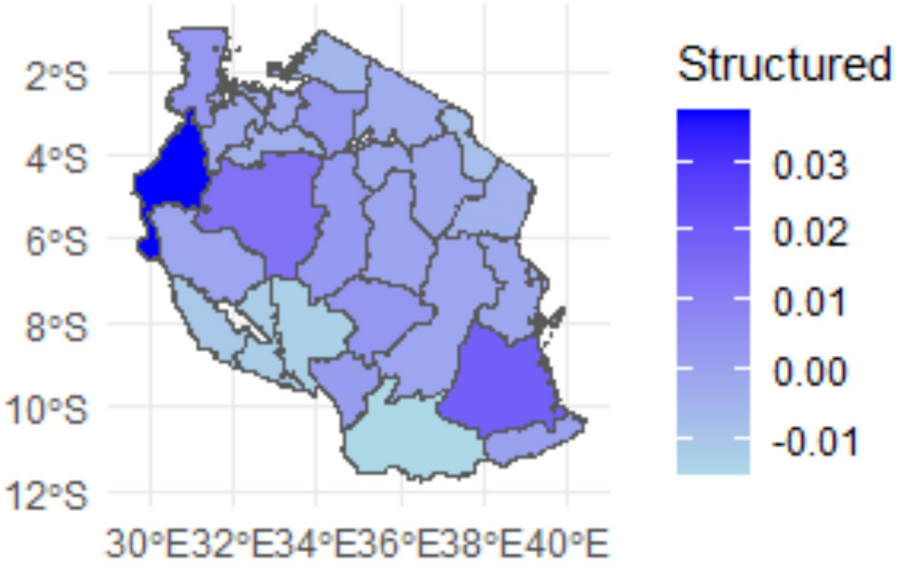



**Fig. 7** Structured spatial variation of CVD posterior estimates in Tanzania mainland

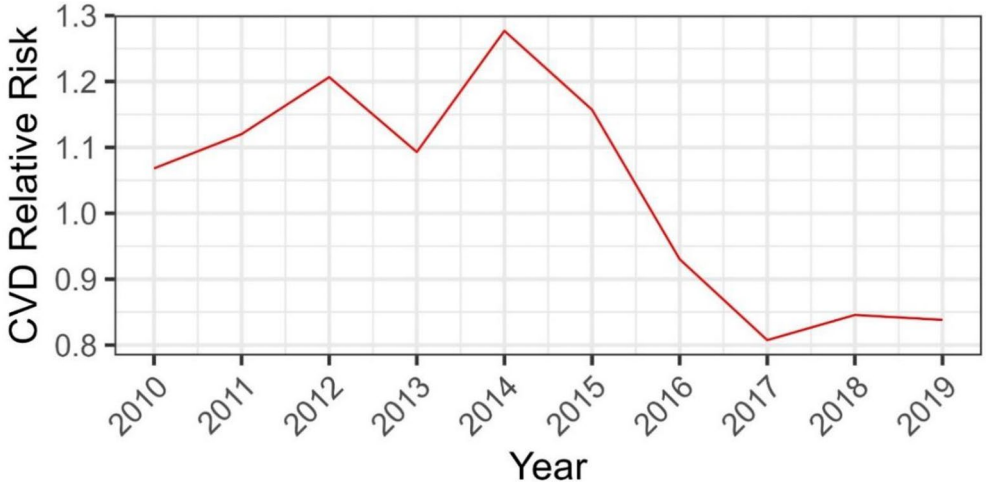



**Fig. 8** Relative risks for temporal effects of CVD (Exponents of Posterior Means), 2010–2019

Fourthly, the spatial-temporal model using INLA method was employed. INLA is a specialized algorithm for the class of Latent Gaussian Model (LGM). The LGM approach tends to adjust the non-spatial and spatial random effect model. The spatial random effect model accounts for geo-reference data, which borrows strength from neighboring regions and the entire geographical region to alleviate estimates based on small, local sample sizes within the areas. INLA was used in this study due to its ability to provide a faster and more accurate estimates of the posterior marginals compared to simulation-based Markov Chain Monte Carlo (MCMC) techniques, which use deterministic nested Laplace approximations.

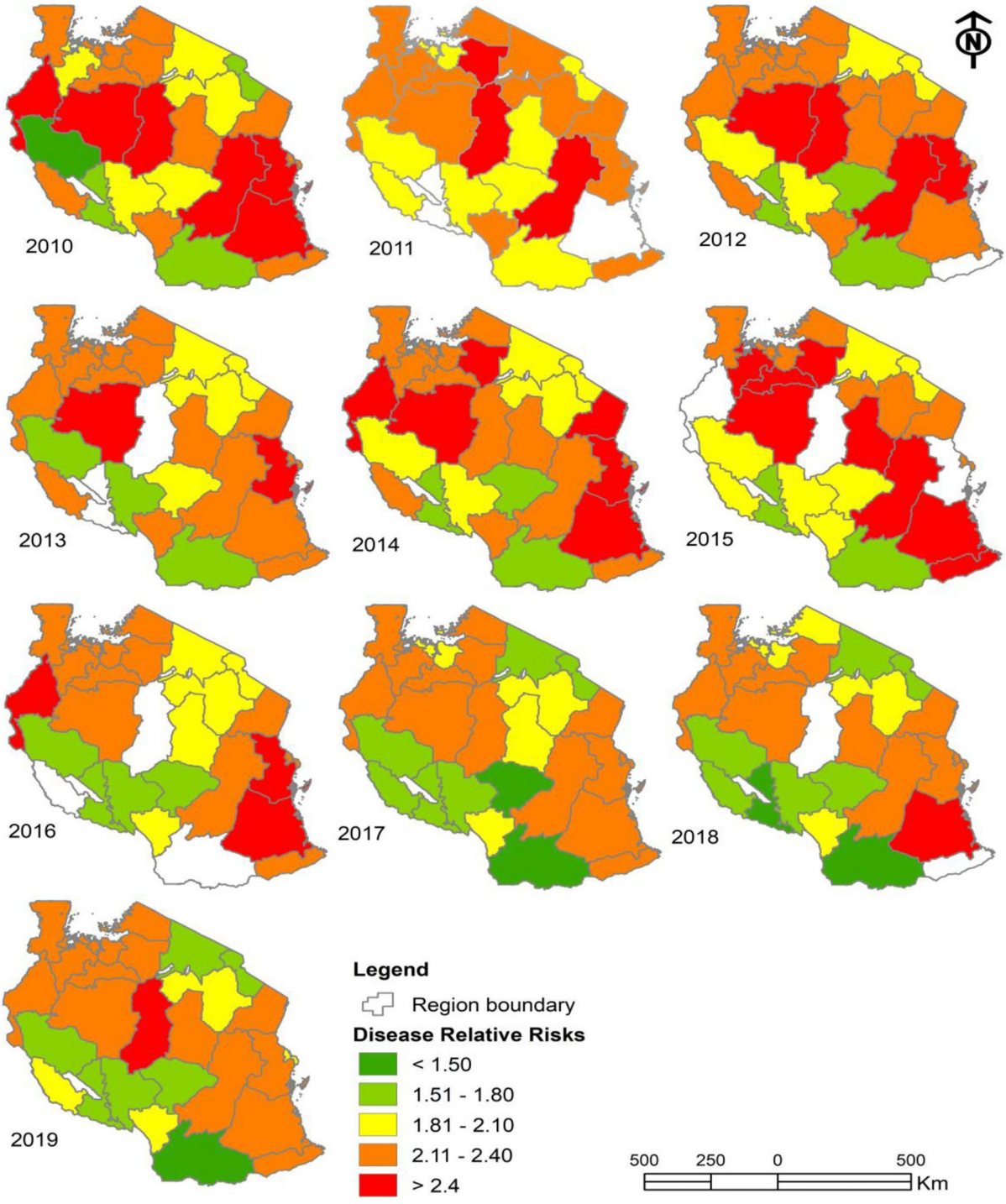



**Fig. 9** Spatiotemporal distribution of CVD incidence rates from 2010 to 2019 in the Tanzania mainland

## Conclusion

This study used Bayesian spatial-temporal modeling to analyze CVD incidence data collected from four tertiary hospitals and the TMA in Tanzania mainland. The model with spatial and temporal random components best predicted the risk of CVD incidence. The spatial and temporal models best fit the CVD incidence data and can be used to predict the risk of CVD incidence. The results show a decrease in CVD incidence from 2015 to 2019. The CVD incidences occurred more in Tanzania's coastal and lake areas between 2010 and 2019. It was also revealed that people aged 60–64 years were more likely to suffer from CVD incidence than those aged 43–59

years. Females, smokers, alcohol drinkers and those with a high BMI are also at high risk of developing CVD. The average annual maximum air temperature was positively correlated with CVD incidence. In this case, the government must educate people to minimize disease risk and uplift the nation's workforce.

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**Author contributions**
B.S and A.M wrote the main document text and M.M cross check the analysis and findings. All authors reviewed the manuscript.

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**Data availability**
The data used during this study are available from the corresponding author for a reasonable request.

## Declarations

**Ethical approval and consent to participate**
Not applicable.

**Consent of publication**
Not applicable.

**Competing interests**
The authors declare no competing interests.

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